12 Answer: C

C is incorrect because to keep moving in a straight line in path C, the spacecraft needs additional force with a radial component to act against pull of Earth's gravity.

This force needs to be adjusted in a way so as to be always equal and opposite to Earth's pull, such that it counteracts the radial acceleration.

Path B would not need any force from rocket if the gravitational force acting on spacecraft is just sufficient to provide the centripetal force to maintain the circular motion.

Paths A and D are paths of free fall when rockets are switched off; D on the way up, A on the way down.